

# AlphaDropout#

<https://pytorch.org/docs/stable/generated/torch.nn.modules.dropout.AlphaDropout>

## Description:

Applies Alpha Dropout over the input. Alpha Dropout is a type of Dropout that maintains the self-normalizing property. For an input with zero mean and unit standard deviation, the output of Alpha Dropout maintains the original mean and standard deviation of the input. Alpha Dropout goes hand-in-hand with SELU activation function, which ensures that the outputs have zero mean and unit standard deviation. During training, it randomly masks some of the elements of the input tensor with probability  $p$  using samples from a bernoulli distribution. The elements to masked are randomized on every forward call, and scaled and shifted to maintain zero mean and unit standard deviation. During evaluation the module simply computes an identity function. More details can be found in the paper Self-Normalizing Neural Networks .

## Function Signature

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```
class torch.nn.modules.dropout.AlphaDropout(p=0.5,  
inplace=False)[source]#
```

Parameters

Shape:

```
forward(input)[source]#
```

Return type

**Returns:**

Tensor

## Code Examples

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```
>>> m = nn.AlphaDropout(p=0.2)  
>>> input = torch.randn(20, 16)  
>>> output = m(input)
```

# Detailed Documentation

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## Documentation

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During training, it randomly masks some of the elements of the input tensor with probability  $p$  using samples from a bernoulli distribution. The elements to masked are randomized on every forward call, and scaled and shifted to maintain zero mean and unit standard deviation.

During evaluation the module simply computes an identity function.

More details can be found in the paper [Self-Normalizing Neural Networks](#) .

$p$  (float) – probability of an element to be dropped. Default: 0.5

`inplace` (bool, optional) – If set to True, will do this operation in-place

Input:  $(*)^{(*)} (*)$ . Input can be of any shape

Output:  $(*)^{(*)} (*)$ . Output is of the same shape as input

Runs the forward pass.

